

INFO8006 Introduction to Artificial Intelligence

Exercise session 2: Games and adversarial search

RECAP

Games

A *game* is the search problem of lecture 2, with two components added and one replaced:

- the *states* $s \in \mathcal{S}$ and the *initial state* $s_1 \in \mathcal{S}$, as before;
- the *actions* $\text{actions}(s) \subseteq \mathcal{A}$ and the *transition model* $s' = \text{result}(s, a)$, as before;
- *new*: a function $\text{player}(s) \in \{1, \dots, N\}$ that says whose turn it is in state s ;
- *replaced*: the goal states G become the *terminal states* $T \subseteq \mathcal{S}$, tested by $\text{terminal-test}(s)$, i.e. $s \in T$;
- *new*: the path cost becomes a *utility* $\text{utility}(s, p)$, the payoff of player p when the game ends in $s \in T$, e.g. 1, 0 or $\frac{1}{2}$ for a win, a loss or a draw.

Together, s_1 , actions and result define the *game tree*, whose nodes are states and whose edges are actions. The solution is a *strategy* $\pi_p : \mathcal{S} \rightarrow \mathcal{A}$, a move for every state the opponent can lead to.

Zero-sum games

In a *zero-sum* game, the total payoff to all players is constant. We assume a deterministic, turn-taking, two-player zero-sum game with perfect information, whose players MAX and MIN maximize and minimize the same utility; MAX moves first.

Minimax

The *minimax value* of s is the largest payoff MAX can achieve against an optimal MIN,

$$\text{minimax}(s) = \begin{cases} \text{utility}(s) & \text{if } s \in T \\ \max_a \text{minimax}(\text{result}(s, a)) & \text{if MAX plays} \\ \min_a \text{minimax}(\text{result}(s, a)) & \text{if MIN plays,} \end{cases}$$

over $a \in \text{actions}(s)$, and the optimal move is $\pi^*(s) = \arg \max_a \text{minimax}(\text{result}(s, a))$. Time $O(b^m)$, space $O(bm)$.

α - β pruning

α and β are the best values found so far for MAX and MIN along the path from the root. A MIN node stops expanding its children as soon as its value $v \leq \alpha$, a MAX node as soon as $v \geq \beta$. Pruning leaves the minimax values unchanged; with a perfect ordering of the moves, it examines $O(b^{m/2})$ nodes.

H-Minimax

Under time constraints, the utility is replaced by an *evaluation function* $\text{eval}(s)$ and the terminal test by a *cutoff test* $\text{cutoff-test}(s, d)$ at depth d ,

$$\text{h-minimax}(s, d) = \begin{cases} \text{eval}(s) & \text{if cutoff} \\ \max_a \text{h-minimax}(\text{result}(s, a), d + 1) & \text{if MAX plays} \\ \min_a \text{h-minimax}(\text{result}(s, a), d + 1) & \text{if MIN plays.} \end{cases}$$

Ideally, eval orders the states as their minimax values do. The cutoff applies to *quiescent* states; consequences beyond the depth limit are hidden by the *horizon effect*.

Expectiminimax

In *stochastic games*, *chance nodes* map a state-action pair to its outcomes r , with probabilities $P(r)$. Writing $v(s) = \text{expectiminimax}(s)$,

$$v(s) = \begin{cases} \text{utility}(s) & \text{if } s \in T \\ \max_a v(\text{result}(s, a)) & \text{if MAX plays} \\ \min_a v(\text{result}(s, a)) & \text{if MIN plays} \\ \sum_r P(r) v(\text{result}(s, r)) & \text{if CHANCE plays.} \end{cases}$$

Monte Carlo tree search

Each node n keeps the number of wins $Q(n, p)$ of player p over the random playouts through n , and its number of visits $N(n)$. A round does *selection*, *expansion*, *simulation* and *backpropagation*; selection picks the child n' of n that maximizes

$$\frac{Q(n', p)}{N(n')} + c \sqrt{\frac{\log N(n)}{N(n')}}.$$

In session: exercises 1, 2.

Exercise 1 Tic-tac-toe (AIMA ch. 5)

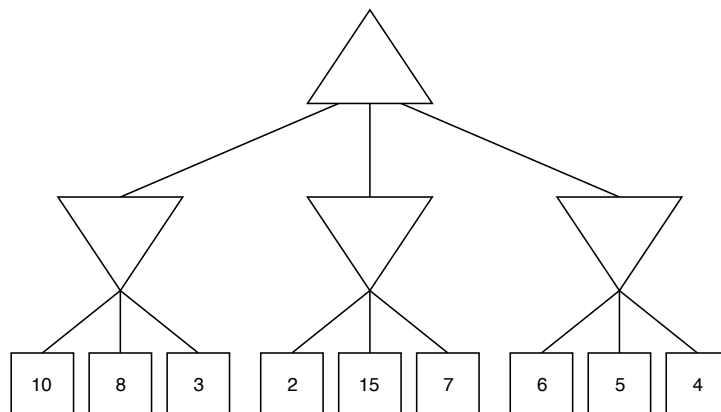
Tic-Tac-Toe is a game for two players, X and O, who take turns marking the cells of a 3×3 grid. The player who succeeds in placing three of their marks in a straight line (horizontal, vertical or diagonal) wins the game. If neither of the players wins before the grid is full, it is a draw.

We consider X as the max player and O as the min player. We define X_n as the number of rows, columns or diagonals with exactly n X's and no O's. Similarly, O_n is the number of rows, columns, or diagonals with just n O's. A position s is terminal if $X_3(s) \geq 1$, $O_3(s) \geq 1$ or if the grid is full. The utility function assigns +1, -1 or 0 to such position, respectively. For non-terminal positions, we use an evaluation function defined as $\text{eval}(s) = 3X_2(s) + X_1(s) - 3O_2(s) - O_1(s)$.

- Define the search problem associated with the Tic-Tac-Toe game.
- Approximately how many possible game states of Tic-Tac-Toe are there?
- Show the whole game tree starting from an empty grid down to depth 2 (one X and one O on the board), taking symmetry into account.
- Annotate your tree with the evaluations of all the positions at depth 2.
- Using the H-Minimax algorithm, annotate your tree with the backed-up values for the positions at depths 1 and 0, and use those values to choose the optimal starting move.
- Is this evaluation function a good heuristic? If not, provide one or more states s for which $\text{eval}(s)$ is misleading.

Exercise 2 Minimax (CS188, Fall 2019)

Consider the zero-sum game tree below. Triangles that point up, such as at the top node (root), represent choices for the maximizing player; triangles that point down represent choices for the minimizing player.



- Assuming both players act optimally, fill in the Minimax value of each node.
- Which nodes can be pruned from the game tree above through α - β pruning? If no nodes can be pruned, explain why not. Assume the search goes from left to right; when choosing which child to visit first, choose the left-most unvisited child.
- Again, consider the same zero-sum game tree, except that now, instead of a minimizing player, we have a chance node that will select one of the three values uniformly at random. Fill in the expectiminimax value of each node. The game tree is redrawn below for your convenience.

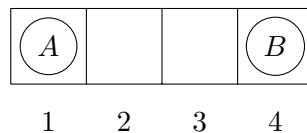
Exercise 3 21 misery game (Exam, January 2019)

The game “21” is played with any number of players who take turns increasing a counter. The counter starts at 1 and each player in turn increases the counter by 1, 2, or 3, but may not exceed 21; the player who says “21” or larger loses.

- Define the search problem associated with the 2-player version of the “21” game.
- For the following, consider the game of “5” (still in its 2-player version), which has the same rules as “21” except that you should not say 5 or more. Show the whole game tree.
- Using the Minimax algorithm, annotate your tree with the backed-up values, and use those values to choose the optimal starting move.

Exercise 4 Leapfrog (AIMA ch. 5)

Consider the following two-player turn-taking game which initial configuration is shown in the figure below. Player *A* moves first. Each player must move their token to an adjacent free cell in either direction. If the opponent occupies an adjacent cell, then a player may jump over the opponent to the next free cell, if any. For example, if *A* is on 3 and *B* is on 2, then *A* may move back to 1. The game ends when a player reaches the opposite end of the board. If player *A* reaches cell 4 first, then the value of the game to *A* is +1; if player *B* reaches cell 1 first, then the value of the game to *A* is -1.



- Define the search problem associated with this game.
- Draw the complete game tree, using the following conventions:
 - Put each terminal state in a square box and annotate it with its game value.
 - Put loop states (states that already appear on the path to the root) in double square boxes. Since their value is unclear, annotate them with a “?” symbol.
- Explain why the standard minimax algorithm would fail on this game.
- Annotate each node with its backed-up minimax value. Explain how you handled the “?” values and why.
- This 4-cell game can be generalized to n cells for any $n > 2$. Prove that *A* wins if n is even and loses if n is odd.

Exercise 5 Chess and transposition table (AIMA ch. 5)

Suppose you have a chess program that can evaluate 16 million nodes per second.

- Decide on a compact representation of a game state for storage in a transposition table.
- About how many entries can you fit in a 4 GB in-memory table?
- Will that be enough for the three minutes of search allocated for one move?
- How many table lookups can you do in the time it would take to do one evaluation? Suppose that you have a 3.2 GHz machine and that it takes 20 operations to do one lookup on the transposition table.

QUIZ

1. In Monte Carlo Tree Search (MCTS), in the formula

$$\frac{Q(n', p)}{N(n')} + c \sqrt{\frac{\log N(n)}{N(n')}},$$

which of the following is true?

- ☐ The first term encourages the exploitation of higher-reward nodes, while the second encourages the exploration of less-visited nodes.
- ☐ The first term encourages the exploration of less-visited nodes, while the second term encourages the exploitation of higher-reward nodes.
- ☐ The first term encourages the exploitation of highly-visited nodes, while the second term encourages the exploration of lesser-rewarding nodes.
- ☐ The first term encourages the exploration of lesser-rewarding nodes, while the second term encourages the exploitation of highly-visited nodes.

2. In adversarial search,

- ☐ The horizon effect arises when the search is stuck in a cycle.
- ☐ The horizon effect arises when the evaluation function is perfect.
- ☐ The deeper in the tree the evaluation function is buried, the more the quality of the evaluation matters.
- ☐ If not looked deep enough, bad moves may appear as good moves, because their consequences are hidden beyond the search horizon.

3. A quiescent state is

- ☐ A state in which the game will loop indefinitely.
- ☐ A state in which an agent is stuck.
- ☐ A state in which the outcome of a game is unlikely to vary a lot in the near future.
- ☐ A state in which the MAX agent is certain to win.

4. Minimax algorithm

- ☐ Cannot fail more against a sub-optimal agent than an optimal one.
- ☐ Can fail more often if the opponent is sub-optimal but predictable and that we underestimate the opponent strategy.
- ☐ Does not ensure that the MAX agent always wins.
- ☐ All of the above.